FREEDOM INTERNATIONAL SCHOOL

WORKSHEET- MCQ (SOLUTIONS)

PHYSICS

WAVES

CLASS XI

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1. The speed of a wave in a medium is 760 m/s. If 3600 waves are passing through a point in the medium in 2 min, then its wavelength is

   (a) 13.8 m (b) 41.5 m (c) 25.3 m (d) 57.2 m

   Ans: (c) 25.3 m

   $v = 760 m/s$

   $n = 3600$

   $t = 2 \times 60 sec = 120 sec$

   $f = \frac{n}{time} = \frac{3600}{2 \times 60} = 30 Hz$

   The relation between frequency f, wavelength λ and speed v is given by

   $v = \lambda f$

   $\lambda = \frac{v}{f}$

   $\lambda = \frac{760}{30} = 25.3 m$

2. The speed of a transverse wave going on a wire having length 50 cm and mass 5.0 g is 80 m/s. The tension developed in the wire will be

   (a) 64 N (b) 44 N (c) 54 N (d) 34 N

   Ans: (a) 64 N

   The linear mass density is $\mu = m/L = \frac{5 \times 10^{-3} kg}{50 \times 10^{-2} m} = 10^{-2} kg\,m^{-1}$

   The wave speed is $v = \sqrt{\frac{T}{\mu}}$

   Tension, $T = \mu v^2 = 10^{-2} \times 6400$

   $= 64 N$

3. If at same temperature and pressure, the densities for two diatomic gases are respectively d_1 and d_2, then the ratio of velocities of sound in these gases will be

(a) $\sqrt{\frac{d_2}{d_1}}$ (b) $d_1 d_2$ (c) $\sqrt{\frac{d_1}{d_2}}$ (d) $\sqrt{d_1 d_2}$

Ans: (a) $\sqrt{\frac{d_2}{d_1}}$

$$v \propto \frac{1}{\sqrt{\rho}} \Rightarrow \frac{v_1}{v_2} = \sqrt{\frac{d_2}{d_1}}$$

4. If the equation of a transverse wave is $y = 5 \sin 2\pi \left(\frac{t}{0.04} - \frac{x}{40}\right)$

where distance is in cm and time in seconds, then the wavelength of wave will be

(a) 20 cm (b) 40 cm (c) 60 cm (d) 80 cm

Ans: (b) 40 cm

$$y = \sin 2\pi \left[\frac{t}{0.04} - \frac{x}{40}\right]$$

$$k = \frac{2\pi}{40}$$

$$\frac{2\pi}{\lambda} = \frac{2\pi}{40} \quad \therefore \lambda = 40cm$$

5. The equation of a wave travelling on a string is

$y = 4 \sin \left[\frac{\pi}{2}\left(8t - \frac{x}{8}\right)\right]$, where x, y are in cm and t in second. The velocity of the wave is

(a) 64 cm/s, in -x direction (b) 32 cm/s, in -x direction

(c) 32 cm/s, in +x direction (d) 64 cm/s, in +x direction

Ans: (d) 64 cm/s, in +x direction

$$\text{Velocity } v = \frac{\omega}{k} = 4\pi / (\pi/16)$$

$$= 64 \text{ cm/s in } +x \text{ direction.}$$

6. The first overtone of a stretched wire of given length is 320 Hz. The first harmonic is

(a) 320 Hz (b) 160 Hz (c) 480 Hz (d) 640 Hz

Ans: (b) 160 Hz

7. For beats to be produced

(a) frequency of sources should be different and amplitude should be same

(b) frequency of sources should be same and amplitude should be different

(c) frequency of sources should be different and amplitude should be different

(d) frequency of sources should be same and amplitude should be same

Ans: (a) frequency of sources should be different and amplitude should be same

8. Following two wave trains are approaching each other: y_1= a sin 200πt, y_2= a sin 208 πt. The number of beats heard per second is

   (a) 8 (b) 4 (c) 1 (d) zero

   Ans:(b) 4

   $\omega_1 = 200\pi$, $\omega_2 = 208\pi$,

   $\omega = 2\pi\upsilon$

   υ_1=100 Hz, υ_2=104 Hz

   $\upsilon_{beat} = \upsilon_2 - \upsilon_1 = 4$ Hz

9. The velocity of sound in open ended tube is 330 m/s, the frequency of wave is 1.1 kHz and length of tube is 30 cm, then number of harmonics that it will emit is

   (a) 2 (b) 3 (c) 4 (d) 5

   Ans: (a) 2

   vn=nv/2L

   1.1×10^3=(n×330)/(2×0.30)

   n=2.

10. A tube closed at one end containing air produces fundamental note of frequency 512 Hz. If the tube is open at both the ends, the fundamental frequency will be

    (a) 256 Hz (b) 768 Hz (c) 1024 Hz (d) 1280 Hz

    Ans: (c) 1024 Hz

    The fundamental frequency for a tube closed at one end is

    $$v_c = \frac{v}{4l}$$

    $$512 \times 4 = \frac{v}{l}$$

    Now, The fundamental frequency for a tube open at both the ends

    $$v_o = \frac{v}{2l}$$

    $$v_o = \frac{1}{2}512 \times 4$$

    $$v_o = 1024\text{Hz}$$

For questions 11 to 15, two statements are given- one labelled Assertion (A) and the other labelled Reason (R). Select the correct answer to these questions from the options as given below.

A. Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
B. Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
C. Assertion is true but Reason is false.

D. Both Assertion and Reason are false.

11. **Assertion:** When two vibrating tuning forks having frequencies 256 Hz and 512 Hz are held near each other, beats cannot be heard.
    Reason: The principle of superposition is valid if the frequencies of the oscillators are nearly equal.
    Ans: C
12. **Assertion:** The fundamental frequency of an open organ pipe increases as the temperature is increased.
    Reason: This is because as the temperature increases, the velocity of sound increases more rapidly than the length of the pipe.
    Ans: A
13. **Assertion:** In a string wave, during reflection from fixed boundary, the reflected wave is inverted.
    Reason: The force on a string by clamp is in downward direction while string is pulling the clamp in the upward direction.
    Ans: A
14. **Assertion:** The velocity of sound increases with increase in humidity.
    Reason: Velocity of sound does not depend upon the medium.
    Ans: C
15. **Assertion:** Solids can support both longitudinal and transverse waves but only longitudinal waves can propagate in gases.
    Reason: For the propagation of transverse waves, medium must also necessarily have the property of rigidity.
    Ans: A